

Endpoint Forensics and Monitoring

Velociraptor | Splunk Enterprise

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Why I Built This

This project pairs two industry-standard endpoint tools to cover both sides of endpoint detection and response: Velociraptor for live forensics and artifact collection, and Splunk Enterprise for log ingestion and SPL-based threat hunting.

The environment was a local Windows 10 host running the Velociraptor server, a remote AWS EC2 instance running Windows Server 2022 as the managed endpoint, and a Splunk instance receiving forwarded Windows event logs via the Universal Forwarder. The work spanned four areas: reviewing Windows Security events natively in Event Viewer, deploying and enrolling an endpoint into Velociraptor, running artifact collections and live VQL queries, and building progressive SPL queries to hunt PowerShell execution and process creation activity.

Component	Detail
Velociraptor Version	0.75.6 (commit 0e657d636)
Velociraptor Server	Taholaptop (Windows 10, localhost:8889)
Managed Endpoint	EC2AMAZ-NGPKETK, Windows Server 2022 Datacenter 21H2
Endpoint Region	AWS us-west-2 (Oregon)
Splunk Version	Enterprise 10.2.2 (localhost:8000)
Splunk Forwarder Target	10.10.0.146:9997

Part 1: Windows Event Log Analysis

Before deploying any tooling, the Windows Event Viewer was used to examine the Security log directly on the local host, which contained 30,768 recorded events at the time of review, establishing a baseline for the telemetry that EDR and SIEM platforms later ingest.

The top event was Event ID 4672 (Special Logon), logged when special privileges are assigned to a new logon session, here for NT AUTHORITY\SYSTEM at system startup. Event ID 4624 (Successful Logon) and a burst of Event ID 5379 (Credential Manager reads) were visible alongside it, also clustered at startup. Recognizing the difference between expected startup noise and an anomalous privilege assignment is a core SOC analyst skill: 4672 is particularly relevant when it appears on an interactive or non-system account, which can indicate privilege escalation.

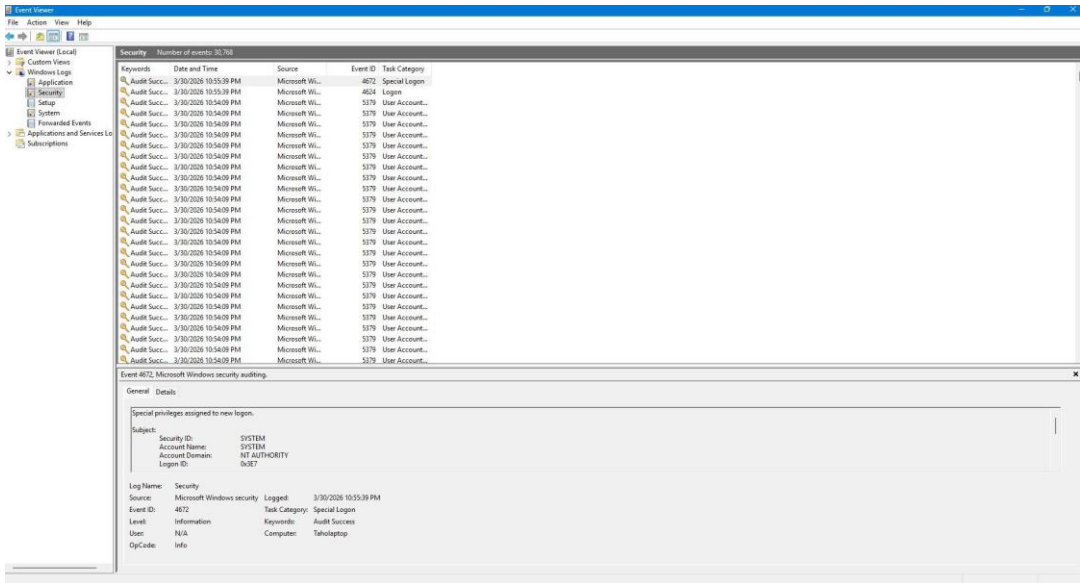


Figure 1. Windows Event Viewer Security log, Event ID 4672 (Special Logon) showing SYSTEM account privilege assignment on Taholaptop.

Events of Interest

- **Event ID 4672, Special Logon:** special privileges assigned to a new logon session; high-value for detecting privilege escalation on non-system accounts.
- **Event ID 4624, Successful Logon:** baseline authentication event, critical for lateral movement detection.
- **Event ID 5379, Credential Manager Read:** a burst at startup is normal; unexpected bursts from unusual processes can indicate credential harvesting.

Part 2: Velociraptor Deployment and Client Enrollment

Server Setup and Dashboard

Velociraptor was deployed in self-signed certificate mode on the local host, with the GUI on localhost:8889 and the client comms endpoint on localhost:9999. The dashboard confirmed a single active organization (root) with the client configuration file ready for download.



The screenshot shows the Velociraptor server dashboard. The 'Current Orgs' section displays a table with one organization named 'root'. The 'Disk Space' section displays a table with disk information for the local fixed disk.

Name	OrgId	ClientConfig
root	root	client.root.config.yaml

DeviceID	Description	VolumeName	VolumeSerialNumber	Size	FreeSpace	Free%
C:	Local Fixed Disk		1CE95F8A	38 GB	15 GB	39

Figure 2. Velociraptor server dashboard, root org created and disk space confirmed.

Endpoint Enrollment

The client configuration YAML, specifying the server URL and embedding the CA certificate for mutual TLS, was deployed to the AWS EC2 endpoint and the Velociraptor client binary was launched. The client overview page confirmed a live green Connected status for EC2AMAZ-NGPKETK with full metadata populated, including OS release, architecture, MAC address, and first/last seen timestamps.

all

EC2AMAZ-NGPKETK.us-west-2.compute.internal

Connected

Overview

Full VQL Drilldown

Shell

Interrogate

VFS

Collected

Client ID

Client Version

Client Build ID

First Seen At

Last Seen At

Last Seen IP

Label

Operating System

Hostname

FQDN

Release

Architecture

MAC Addresses

0-3261940000000000

0.75.6

2020-10-10 10:10:10

2020-04-03 14:30:10

2020-04-03 17:03:50 9712

0:11:40001

Label

Windows

EC2AMAZ-NGPKETK

EC2AMAZ-NGPKETK.us-west-2.compute.internal

Microsoft Windows Server 2019 Datacenter

amd64

08:00:00:00:00:00

Client Metadata

Figure 3. Enrolled endpoint EC2AMAZ-NGPKETK showing Connected status, agent version 0.75.6, and full host metadata.

Part 3: Artifact Collection and VQL Analysis

Hunt: Windows.System.Pslist

A hunt was created against all enrolled clients using the Windows.System.Pslist artifact, which enumerates running processes with PID, PPID, executable path, command-line arguments, username, and Authenticode signature data. The hunt completed in 1 second against the single enrolled endpoint, returning 73 process rows, including standard Windows Server processes (Registry, smss.exe, csrss.exe, wininit.exe, winlogon.exe, services.exe, lsass.exe, svchost.exe) and both Velociraptor agent processes, confirming live agent activity.



Figure 4. Windows.System.Pslist hunt results: raw JSON output showing 73 processes collected from EC2AMAZ-NGPKETK in 1 second.

VQL Notebook: Process Hierarchy Query

A live VQL query was run directly against the connected endpoint through the Velociraptor Notebook, mapping processes to their parents:

```
SELECT Pid, Name, ParentName FROM pslist()
```

The query returned 75 rows and rendered the full Windows process tree: System (PID 4) as root, Registry and smss.exe as its children, wininit.exe spawning services.exe and lsass.exe, and services.exe spawning multiple svchost.exe instances. This parent-child lineage is foundational to detecting process injection, LOLBIN abuse, and other malicious execution patterns that show up as unusual parent-child relationships.

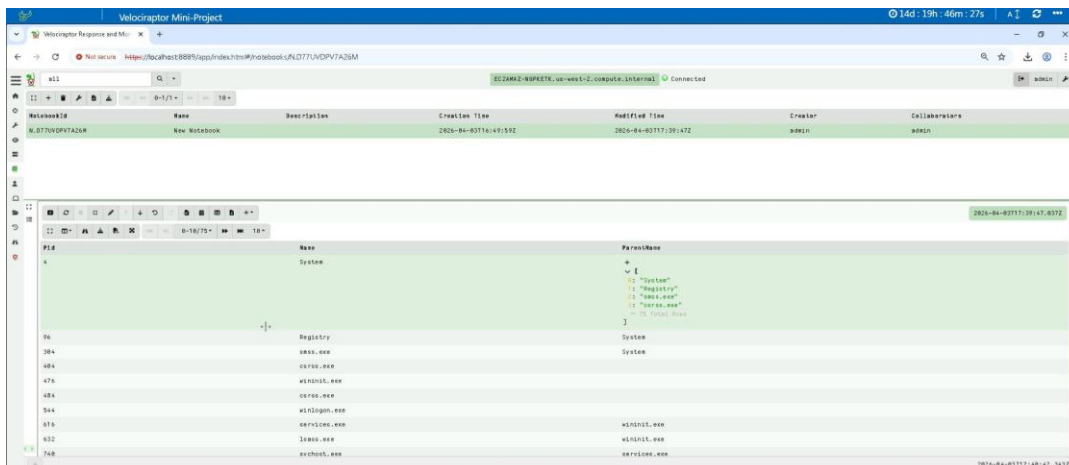


Figure 5. VQL Notebook pslist query: 75-row process hierarchy showing Pid, Name, and ParentName for the enrolled endpoint.

Part 4: Splunk, Log Ingestion and Threat Hunting

Universal Forwarder Setup and Data Verification

The Splunk Universal Forwarder was installed on the EC2 endpoint and configured to forward Windows event logs to the indexer at 10.10.0.146:9997. A broad search confirmed data was flowing:

```
(index=* OR index=_*)
```

The search returned 8,852 events in a 15-minute window, confirming ingestion from host EC2AMAZ-5PNITSP across 34 distinct sources and 25 source types, including Splunk's own audit trail and internal resource usage metrics, confirming the pipeline was live end to end.

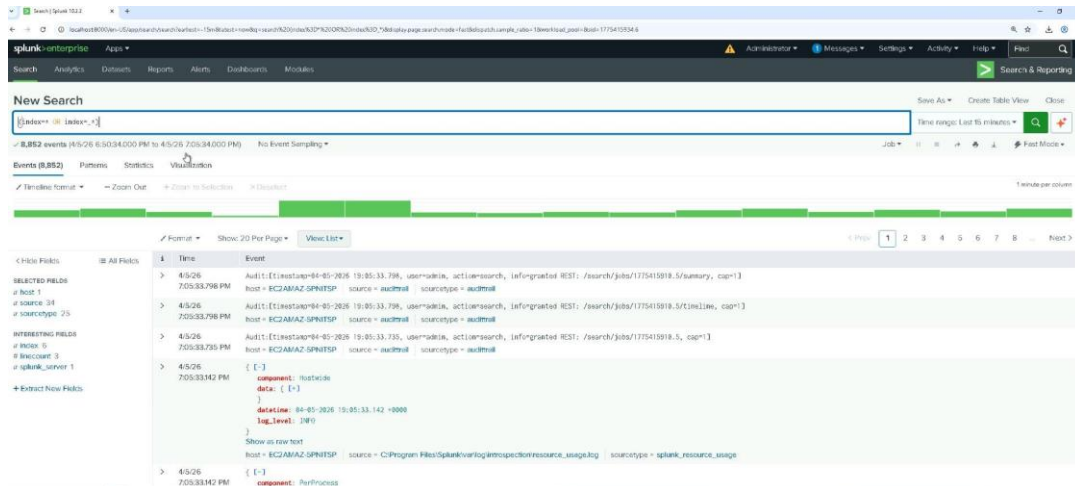


Figure 6. Splunk search returning 8,852 events, confirming log ingestion from EC2AMAZ-5PNITSP.

PowerShell Script Block Logging: Event ID 4104

Event ID 4104 captures the full text of every executed script block, making it one of the most valuable log sources for detecting obfuscated commands, download cradles, and living-off-the-land techniques. The following SPL query extracts the actual commands from the raw event text:

```
sourcetype="WinEventLog:Microsoft-Windows-PowerShell/Operational" EventCode=4104
| rex field=_raw "(?ms)Message=Creating Scriptblock text \((1 of
1\):\\s*(?<cmd>.*?)\\s*ScriptBlock ID:"
| stats count by cmd
| sort - count
```

The query matched 49 events and extracted 10 distinct commands, the highest-frequency being `Get-VM | Select-Object Name, Id, State`, alongside internal runtime blocks and three Splunk Universal Forwarder administrative commands (`bttool inputs list --debug`, `list forward-server`, `list monitor`) run via PowerShell. Their presence confirmed script block logging was capturing real administrative activity, including the forwarder's own configuration, in real time.

Findings Summary

Source	Event / Query	Result	Analytical Value
Event Viewer	EventCode 4672	SYSTEM privilege assignment at startup	Baseline for spotting privilege escalation on non-system accounts
Velociraptor Hunt	Windows.System.Pslist	73 processes in 1 second	Confirms live artifact collection at scale across enrolled endpoints
VQL Notebook	SELECT Pid, Name, ParentName FROM pslist()	75-row process tree	Parent-child lineage needed to catch process injection and LOLBIN abuse
Splunk SPL	EventCode 4104, rex extraction	49 events, 10 unique commands	Full script block text recovers obfuscated or living-off-the-land commands
Splunk SPL	EventCode 4688, timechart	110 events, 91 processes	Reconstructs a precise behavioral timeline from process creation alone

Lessons Learned

1. **Baseline noise before hunting for anomalies.** The startup cluster of 4672, 4624, and 5379 events is routine; without first establishing what normal startup activity looks like, none of it can be judged as anomalous or not.
2. **Live query access changes investigation speed.** The VQL notebook let a process hierarchy query run directly against a remote endpoint and return in seconds, compared to waiting on a scheduled hunt or a forwarded log to land in the SIEM.
3. **Script block logging defeats simple obfuscation.** Event ID 4104 captures the fully resolved command text regardless of how it was invoked, which is what makes it more valuable for hunting than relying on process creation events alone.
4. **Process creation timelines validate administrative activity.** The Splunk worker process burst in the 4688 timechart lined up exactly with when the forwarder was being configured, showing that process-level telemetry can confirm or contradict a claimed timeline of events.
5. **EDR and SIEM answer different questions.** Velociraptor gave point-in-time forensic detail on a single endpoint; Splunk gave a searchable, aggregated view across time. Neither replaces the other in a real detection stack.

Tools and Technologies

Tool	Role
Velociraptor v0.75.6	VQL-based EDR platform for live endpoint forensics and artifact collection
Splunk Enterprise v10.2.2	SIEM for log aggregation, SPL search, and visualization
Splunk Universal Forwarder	Windows agent forwarding event logs to the Splunk indexer
Windows Event Viewer	Native tool for Security, System, and Application log review
AWS EC2	Windows Server 2022 Datacenter 21H2 used as the remote managed endpoint
VQL	Velociraptor Query Language, SQL-like queries for forensic artifacts
SPL	Splunk Processing Language, used for all search, extraction, and visualization

All activity performed in a personal lab environment. April 2026.